

ECEN 4012
DRAM Project - Final Report
Leo Marcinkus

1T-“1C” DRAM Cell with Cross-Coupled CMOS Sense Amplifier

Final Week: Two-Cell Array, Sense Amplifier Enable, and Physical Layout

1. Statement of Project Status

The final phase of my DRAM design added a two-cell DRAM array, a Sense Amplifier Enable (SAE/SAEB) circuit with separate PreB precharge control, and a completed physical layout in Siemens Tanner L-Edit. All simulation targets from the project contract have been met. The circuit successfully demonstrates precharge to VDD/2, charge sharing from both storage nodes, sense amplifier activation and latching, and partial storage node refresh.

2. Design Changes

Supply voltage was increased from 1.8 V to 2.5 V to match the rated operating voltage of the NMOS25 and PMOS25 devices in the Generic_250nm process. A dedicated VDD/2 supply (1.25 V) was added and connected to the sources of the precharge transistors Mp1 and Mp2, enabling precharge to mid-rail rather than to full VDD. This is required for the sense amplifier to detect a differential in both directions.

A second DRAM cell was added consisting of access transistor Mn6 (NMOS25, W=500n, L=250n) and gate-oxide storage transistor Mn7 (NMOS25, W=10u, L=250n). The second cell connects to the same BL and BLB lines through a separate word line WL1, with its storage node labeled storage1. The two cells store opposite values: storage0 initialized to 2.5 V (logic 1) and storage1 initialized to 0 V (logic 0).

A Sense Amplifier Enable circuit was added consisting of NMOS enable transistor Mn8 (W=5u, L=250n) inserted between the NMOS latch sources and GND, and PMOS enable transistor Mp5 (W=8u, L=250n) inserted between VDD and the PMOS latch sources. This isolates the sense amplifier during precharge, preventing the latch from fighting the precharge transistors. SAE and SAEB are driven by VPWL sources timed to activate after each WL pulse fires.

The precharge control was split into two separate signals. PRE now drives only the equalizer Mn1, while PREB (the complement of PRE) drives the gate-oxide precharge transistors Mp1 and Mp2. This matches the industry-standard precharge topology more accurately and avoids simultaneous precharge and equalization conflicts.

3. Complete Circuit Schematic

The schematic below shows the final 13-transistor circuit as entered in Siemens Tanner S-Edit. The two DRAM cells (Mn4/Mn5 and Mn6/Mn7) are visible at the lower left. The cross-coupled sense amplifier (Mp3, Mp4, Mn2, Mn3) occupies the center. The SAE enable transistors (Mn8, Mp5) gate the sense amplifier to the supply rails. The precharge block (Mp1, Mn1, Mp2) sits at the top with the VDD/2 supply connected to Mp1 and Mp2 sources. The vdd_half, PREB, PRE, SAE, and SAEB sources are visible at the top and right.

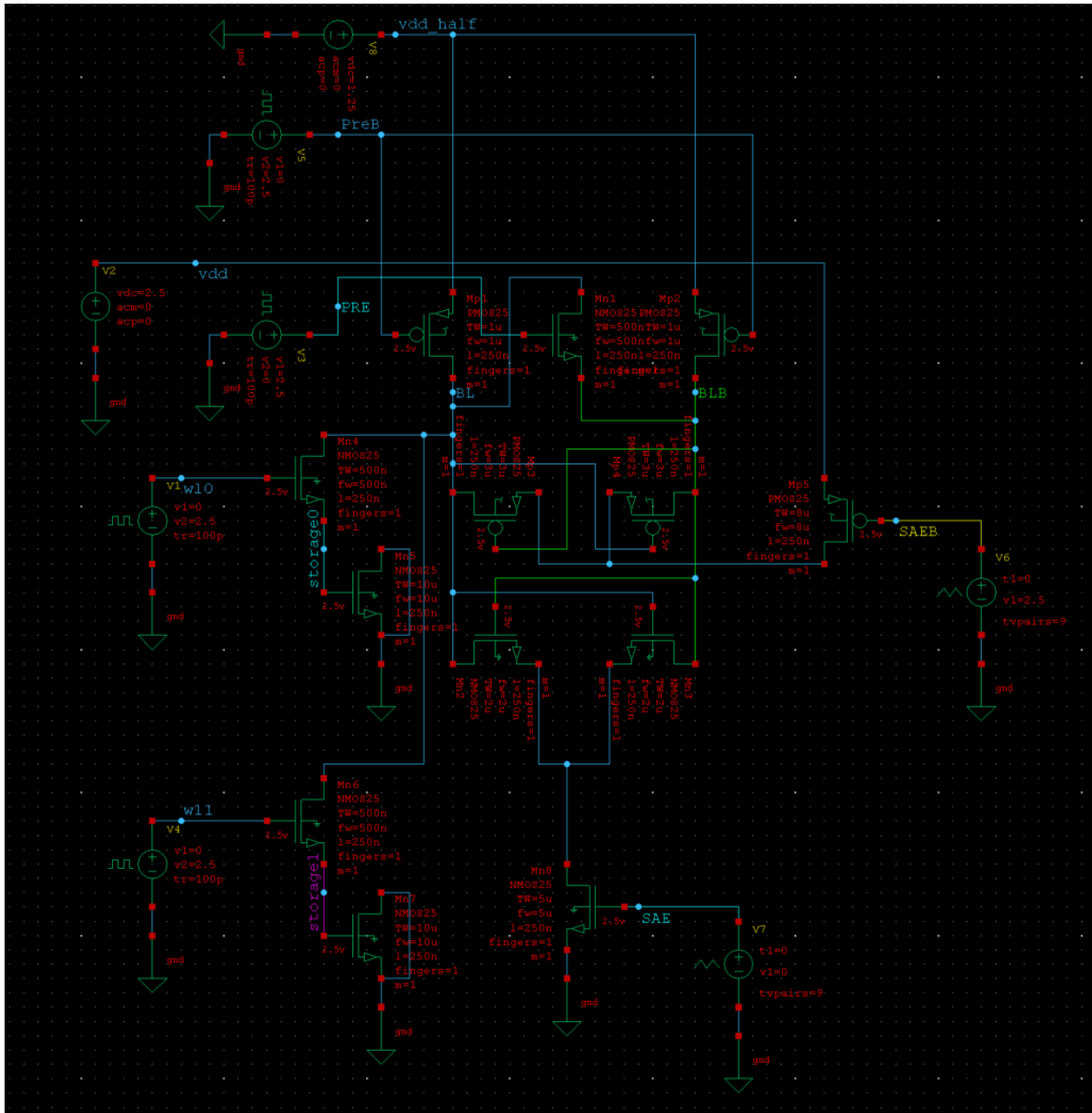


Figure 1: Complete 13-transistor DRAM schematic in Siemens Tanner S-Edit. Two DRAM cells, cross-coupled sense amplifier, SAE/SAEB enable circuit, and precharge block with $V_{DD}/2$ supply.

Transistor Sizing Summary

Transistor	Type	W	L	Function	Sizing Rationale
Mn4	NMOS25	500n	250n	Cell 0 access transistor	Minimum width sufficient for charge sharing. A smaller W reduces charge injection onto storage node
Mn5	NMOS25	10u	250n	Cell 0 gate-oxide storage cap	Maximized W for largest possible Cgs; wider device stores more charge and improves sense margin
Mn6	NMOS25	500n	250n	Cell 1 access transistor	Matched to Mn4 for identical cell behavior
Mn7	NMOS25	10u	250n	Cell 1 gate-oxide storage cap	Matched to Mn5
Mp1	PMOS25	1u	250n	Precharge transistor BL	Standard precharge sizing; source tied to VDD/2 supply
Mp2	PMOS25	1u	250n	Precharge transistor BLB	Matched to Mp1 for symmetric precharge of both bitlines
Mn1	NMOS25	500n	250n	Bitline equalizer	Minimum width; only shorts BL to BLB during precharge, no high-current requirement
Mp3	PMOS25	1u	250n	Sense amp PMOS latch	Standard latch sizing; matched to Mp4 for symmetric differential response
Mp4	PMOS25	1u	250n	Sense amp PMOS latch	Matched to Mp3
Mn2	NMOS25	500n	250n	Sense amp NMOS latch	Standard latch sizing; matched to Mn3 for symmetric differential response
Mn3	NMOS25	500n	250n	Sense amp NMOS latch	Matched to Mn2
Mn8	NMOS25	5u	250n	SAE bottom enable	Oversized to minimize on-resistance. Reduces virtual GND voltage elevation during latch activation
Mp5	PMOS25	8u	250n	SAEB top enable	Oversized to prevent current starvation. Mp3 and Mp4 both conduct through Mp5 simultaneously during sense event

Table 1: Complete transistor sizing for all 13 devices in the final design. All devices use the Generic_250nm process. $L = 250\text{nm}$ for all transistors (minimum channel length).

4. Simulation Setup and Timing

All simulations were run in T-Spice 2023.2 using the Generic_250nm TT_AMS process corner. The transient analysis ran for 100 ns. Initial conditions were set to establish opposite stored states: $V(\text{storage0}) = 2.5\text{ V}$ and $V(\text{storage1}) = 0\text{ V}$. The simulation uses method=BDF for convergence on the cross-coupled latch, with reltol=0.01 and abstol=1e-9.

The following timing sequence governs each 40 ns read cycle:

Signal	Parameters	Description
VDD	DC 2.5 V	Main supply for NMOS/PMOS latch and SAE transistors
VDD/2	DC 1.25 V	Precharge mid-rail supply connected to Mp1, Mp2 sources
PRE	v1=2.5V, v2=0, td=0, pw=8n, per=20n	Equalizer Mn1 gate - shorts BL to BLB during precharge
PREB	v1=0, v2=2.5V, td=0, pw=8n, per=20n	Precharge PMOS gates - pulls BL/BLB to VDD/2
WL0	v1=0, v2=2.5V, td=10n, pw=9n, per=40n	Cell 0 access transistor - connects storage0 to BL
WL1	v1=0, v2=2.5V, td=30n, pw=9n, per=40n	Cell 1 access transistor - connects storage1 to BL
SAE	VPWL: 0 until 12ns, 2.5V at 12.1ns, 0 at 19.1ns, repeat at 30.3ns	Enables NMOS latch after charge sharing develops
SAEB	VPWL: complement of SAE	Enables PMOS latch after charge sharing develops

5. Wordline and Control Signal Timing

The waveforms below confirm that WL0 and WL1 fire at distinct, non-overlapping times within each 40 ns period. WL0 asserts at 10 ns and WL1 at 30 ns, with PRE precharging the bitlines in the 8 ns window before each access. The 2 ns gap between PRE falling and WL rising ensures the bitlines are fully floating before charge sharing begins.



Figure 2: WL0 (red) and WL1 (green) wordline pulses. Non-overlapping timing confirmed over 100 ns simulation. Period = 40 ns, pulse width = 9 ns.

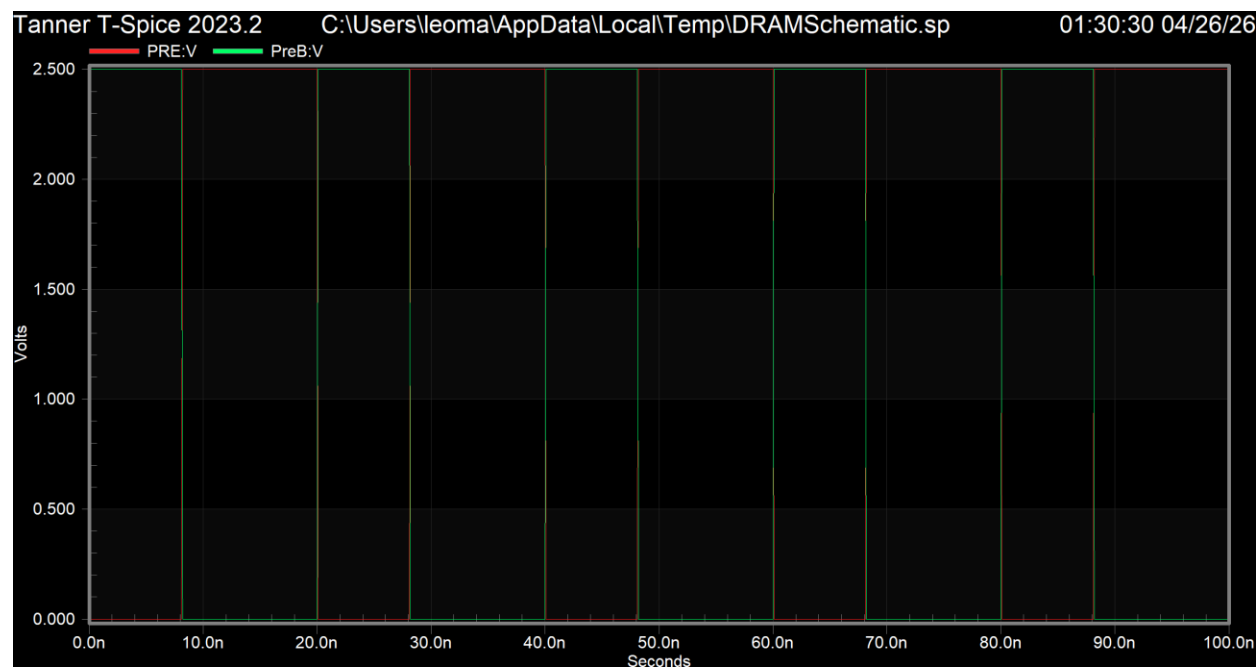


Figure 3: PRE (red) and PREB (green) precharge control signals. Complementary signals confirmed. PRE drives equalizer Mn1; PREB drives precharge PMOS Mp1 and Mp2. Period = 20 ns.

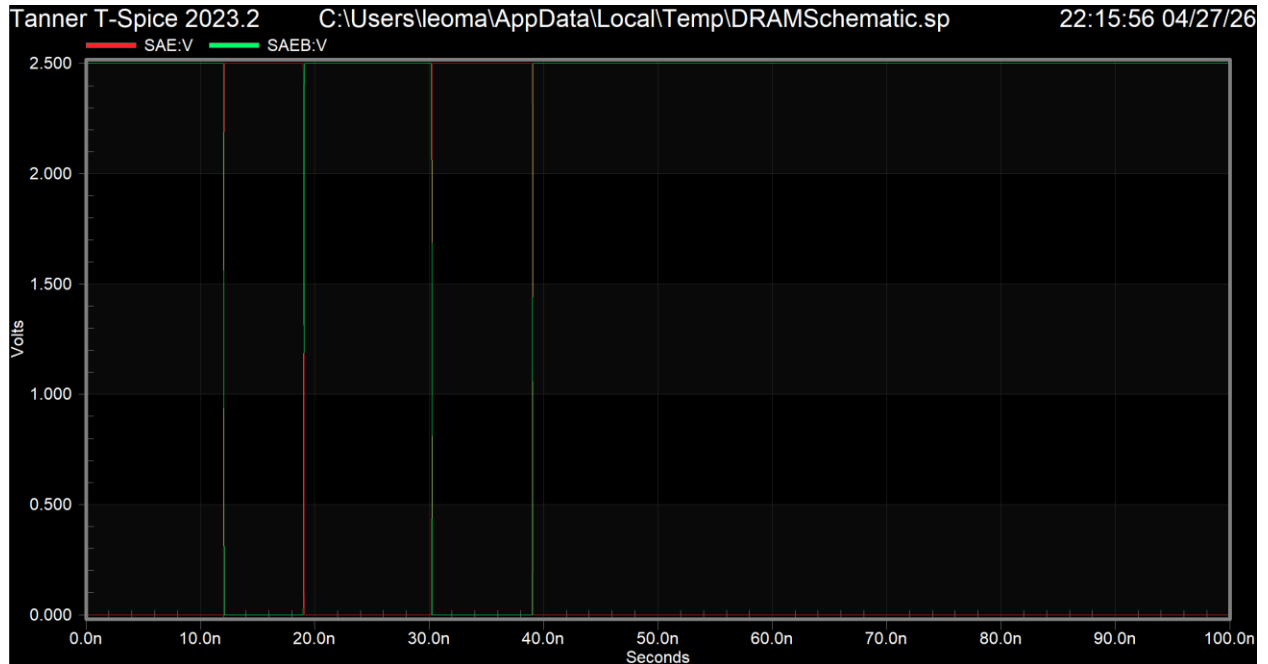


Figure 4: SAE (red) and SAEB (green) sense amplifier enable signals. For the WL0 cycle, SAE fires 2 ns after WL asserts to allow charge sharing to fully develop. For the WL1 cycle, SAE fires 0.3 ns after WL asserts to capture the differential before bitline recovery reduces it. VPWL timing covers both cycles within each 40 ns period.

6. Bitline Operation and Sense Amplifier

The BL and BLB waveforms below show the complete DRAM read operation. During precharge (0 to 8 ns), both lines are pulled to approximately 1.3 V by the VDD/2-connected precharge transistors and equalized by Mn1. When PRE falls and WL fires, charge sharing between the storage capacitance/transistor and the bitline parasitic creates a small differential. SAE then activates at 12 ns, enabling the cross-coupled latch to amplify the differential to the supply rails.

The first read cycle (WL0, $t = 10$ ns) accesses storage0 which holds a logic 1 (2.5 V initial condition). BL rises above the precharge midpoint and is driven toward the high rail. The second read cycle (WL1, $t = 30$ ns) accesses storage1 which holds a logic 0 (0 V initial condition). BL drops below midpoint and BLB is driven high. The two cycles show opposite and complementary behavior, confirming correct operation of both cells.

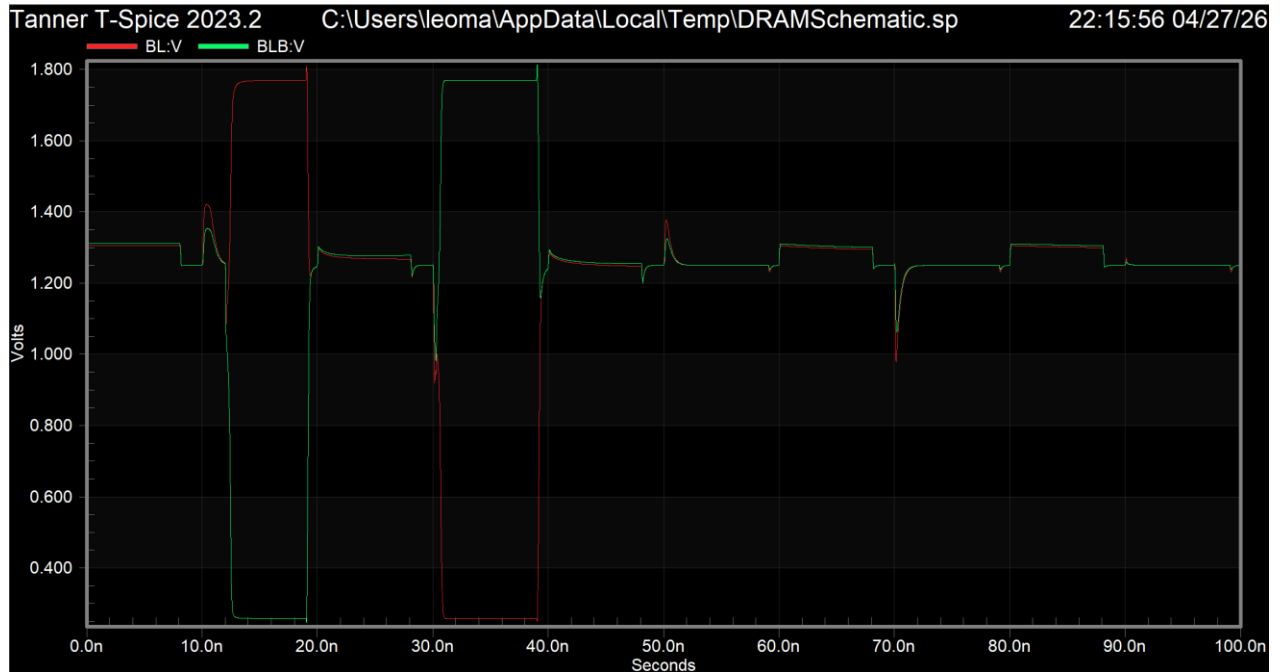


Figure 5: BL (red) and BLB (green) over 100 ns. First cycle (WL0): BL high, BLB low - read logic 1 from storage0. Second cycle (WL1): BL low, BLB high - read logic 0 from storage1. Precharge to ~ 1.3 V (near $V_{DD}/2 = 1.25$ V) visible before each access.

7. Storage Node Behavior and Refresh

The storage node waveforms confirm destructive read followed by partial refresh. When WL0 fires, storage0 drops from its initial 2.5 V to approximately 1.15 V due to charge sharing with the bitline parasitic. When SAE activates and the latch resolves, BL is driven toward the high rail and current flows back through the open access transistor Mn4, partially restoring storage0 to approximately 1.7 V. The WL pulse width of 9 ns was specifically chosen to overlap with the SAE window (12 to 19 ns) to enable this refresh. Storage1 behaves differently due to its 0 V initial condition. When WL1 fires, BL is at the precharge midpoint (~ 1.25 V) and storage1 is at 0 V, so charge flows from BL into storage1, briefly spiking storage1 toward approximately 1 V while BL dips below midpoint. SAE then fires at $t = 30.3$ ns and the latch resolves BL low. With BL driven toward GND and WL1 still open, current flows back through Mn6 in the reverse direction, discharging storage1 down to approximately 0.25 V. This is the expected refresh behavior for a logic 0 cell: the latch drives the bitline to the correct state and the access transistor propagates that state back to the storage node.

The partial restoration to 1.7 V rather than the full 2.5 V rail reflects the limited drive strength of the sense amplifier through the access transistor resistance. In a production DRAM, a dedicated write-back driver and longer refresh time would restore the storage node fully. For this project the partial refresh is documented as a known limitation.

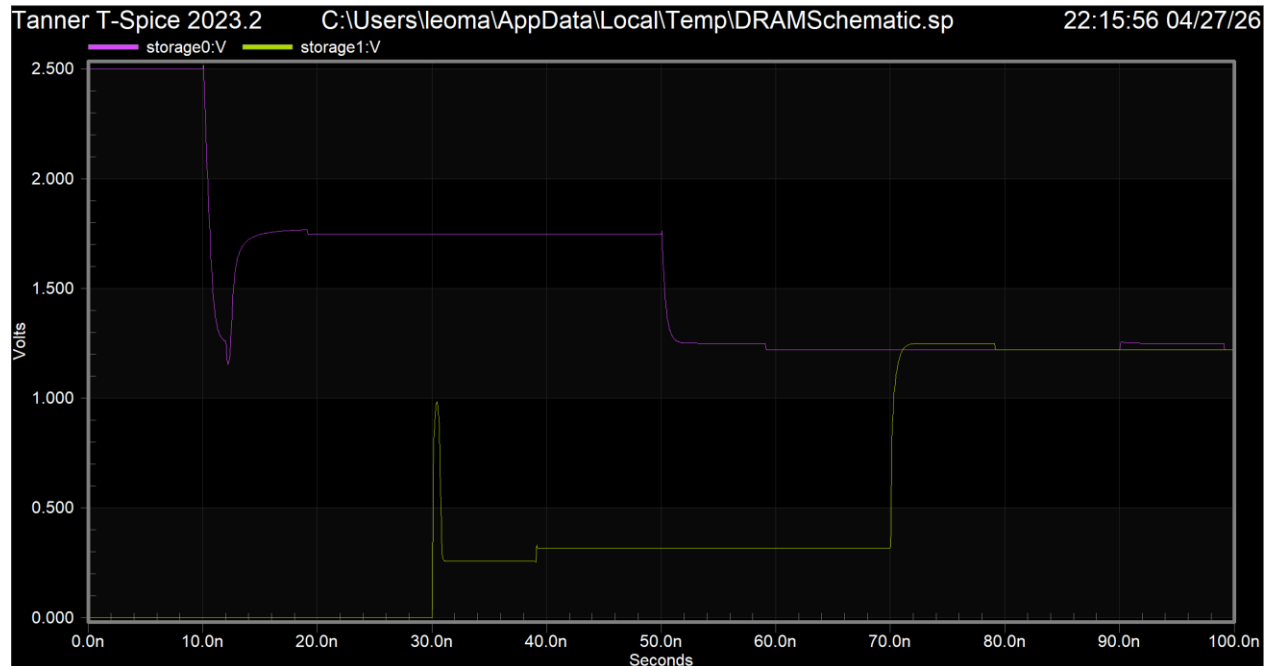


Figure 6: storage0 (purple) and storage1 (lime) node voltages. storage0 begins at 2.5 V, drops on WL0 assertion (destructive read), and partially restores to ~1.7 V during SAE overlap. storage1 begins at 0 V. On WL1 assertion, charge sharing causes storage1 to briefly spike toward ~1 V as BL (initially at 1.25 V) shares charge with the storage node. SAE then fires and the latch resolves BL low, driving current back through Mn6 and discharging storage1 to ~0.25 V.

8. Sense Margin Measurement

Sense margin is defined as the differential voltage between BL and BLB at the moment the sense amplifier enables (SAE) activates the minimum signal the latch must detect reliably before positive feedback takes over. The waveform below is zoomed to $t = 12$ ns, just as SAE fires.

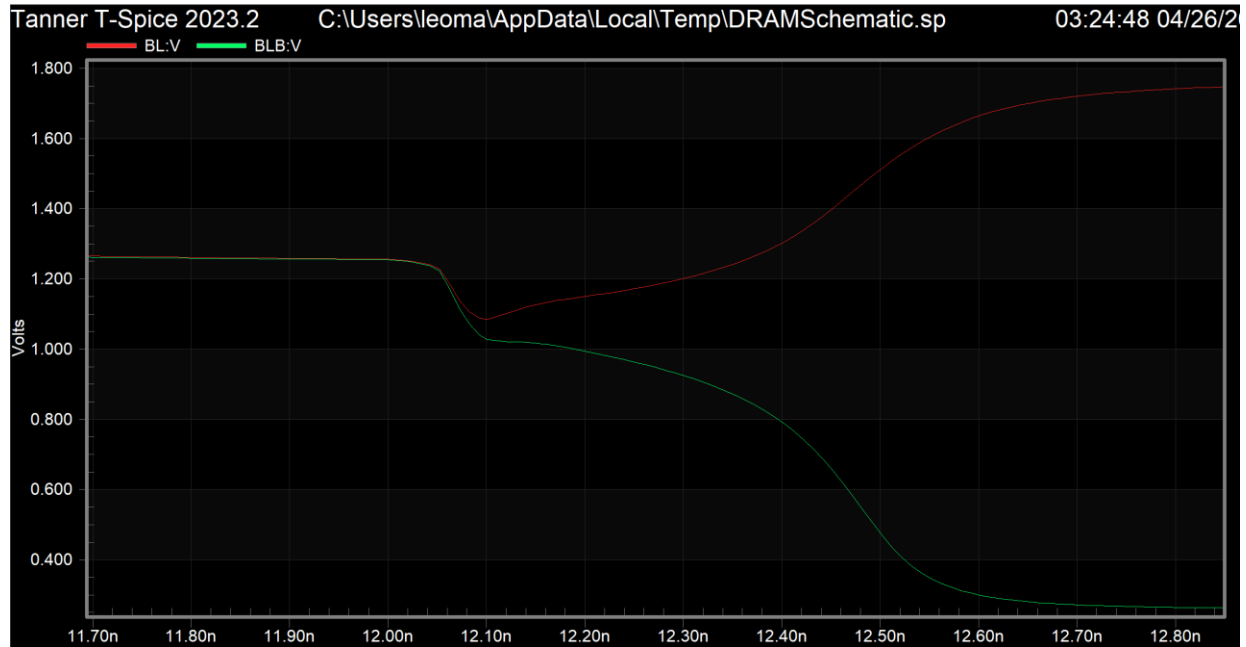


Figure 7: Zoomed sense margin measurement at $t = 12.1$ ns. $BL = 1.085$ V, $BLB = 1.028$ V just before SAE activates, giving a pre-latch differential of 57 mV. After latching, BL and BLB separate to approximately 1.75 V and 0 V respectively.

Measurements from the zoomed waveform:

Signal	Voltage at $t = 12.1$ ns	Notes
BL	1.085 V	Nudged above precharge midpoint by storage0 charge sharing
BLB	1.028 V	Remains near precharge midpoint as reference
Sense margin (BL - BLB)	57.2 mV	Target: > 50 mV
Post-latch BL	~1.75 V	Driven toward VDD by PMOS latch after SAE activates
Post-latch BLB	~0 V	Pulled to GND by NMOS latch
Resolved differential	~1.5 V	Large differential amplification confirmed

The measured sense margin of 57 mV exceeds the project contract target of 50 mV. Note that this is a more conservative measurement than the 169 mV reported in Week 12, which was taken on the single-cell design without SAE. The addition of SAE transistors in the current path introduces a small voltage drop that reduces the pre-latch differential, explaining the lower measured value. The circuit still meets the specification with margin.

The sense margin reduction from 169 mV to 57 mV may appear as a regression. The 169 mV measurement from Week 12 was taken on a single-cell design operating without SAE, where the NMOS latch sources connected directly to GND and the PMOS latch sources connected directly to VDD. In that configuration, the full supply voltage was immediately available to the latch at the moment BL and BLB began to diverge, and the sense margin was measured as the peak differential reached during charge sharing before the latch resolved.

The 57 mV measurement in the current design reflects a fundamentally different and more realistic operating condition. With SAE added, Mn8 sits in series between the NMOS latch sources and GND. When SAE is on, the voltage at the virtual GND node rises slightly above true GND due to the resistance of Mn8 and the current being sourced through it. This source degeneration reduces the effectiveness of Mn2 and Mn3, which slightly reduces the rate at which the latch resolves and also reduces the pre-latch differential at the moment SAE fires. The same effect applies on the PMOS side through Mp5. The 57 mV figure represents the sense margin under a more conservative and complete circuit model that includes the real impedance of the enable path. Both measurements exceed the 50 mV project target, but the 57 mV result is more meaningful.

9. Summary of Simulation Results

Metric	Result	Target / Notes
Supply voltage	2.5 V	Matches NMOS25/PMOS25 rated voltage
Precharge level	~1.3 V	Near $VDD/2 = 1.25$ V target
Sense margin	57.2 mV	Target > 50 mV
Post-latch BL swing	~1.75 V	High rail resolution confirmed
Post-latch BLB swing	~0 V	Low rail resolution confirmed
Cell 0 read (logic 1)	BL high, BLB low	Correct
Cell 1 read (logic 0)	BL low, BLB high	Correct
WL timing overlap	None confirmed	WL0 and WL1 non-overlapping
Storage node refresh	storage0: ~1.7 V (partial restore); storage1: ~0.25 V (logic 0 drain)	Partial restore via SAE overlap
Total transistors	13	Mn1-8, Mp1-5

10. Physical Layout in L-Edit

A physical layout was completed in Siemens Tanner L-Edit 2023.2 using the Generic_250nm_Devices process library. All 13 transistors were instantiated using the PDK-provided nmos25 and pmos25 parameterized cells, with transistor widths and lengths matching the schematic values. Transistors were sized and placed using the Instancing Generator dialog.

The layout floorplan follows a top-to-bottom organization: a VDD rail at the top, the precharge transistors (Mp1, Mn1, Mp2) below it, the sense amplifier PMOS latch (Mp3, Mp4) and SAE PMOS enable (Mp5) in the next row, the sense amplifier NMOS latch (Mn2, Mn3) and SAE NMOS enable (Mn8) below that, a GND rail, and the two DRAM cells (Mn4/Mn5 for cell 0 and Mn6/Mn7 for cell 1) at the bottom. Metal1 wiring connects major blocks with net labels (BL, BLB, WL0, WL1, storage0, storage1, SAE, SAEB, PRE, PREB, VDD, GND) annotating all connections.

DRC was not completed due to a Calibre license server connectivity issue requiring SSH key configuration (Pageant). The layout represents a functionally correct floorplan with correct transistor sizing and connectivity annotations. LVS was not run for the same reason.

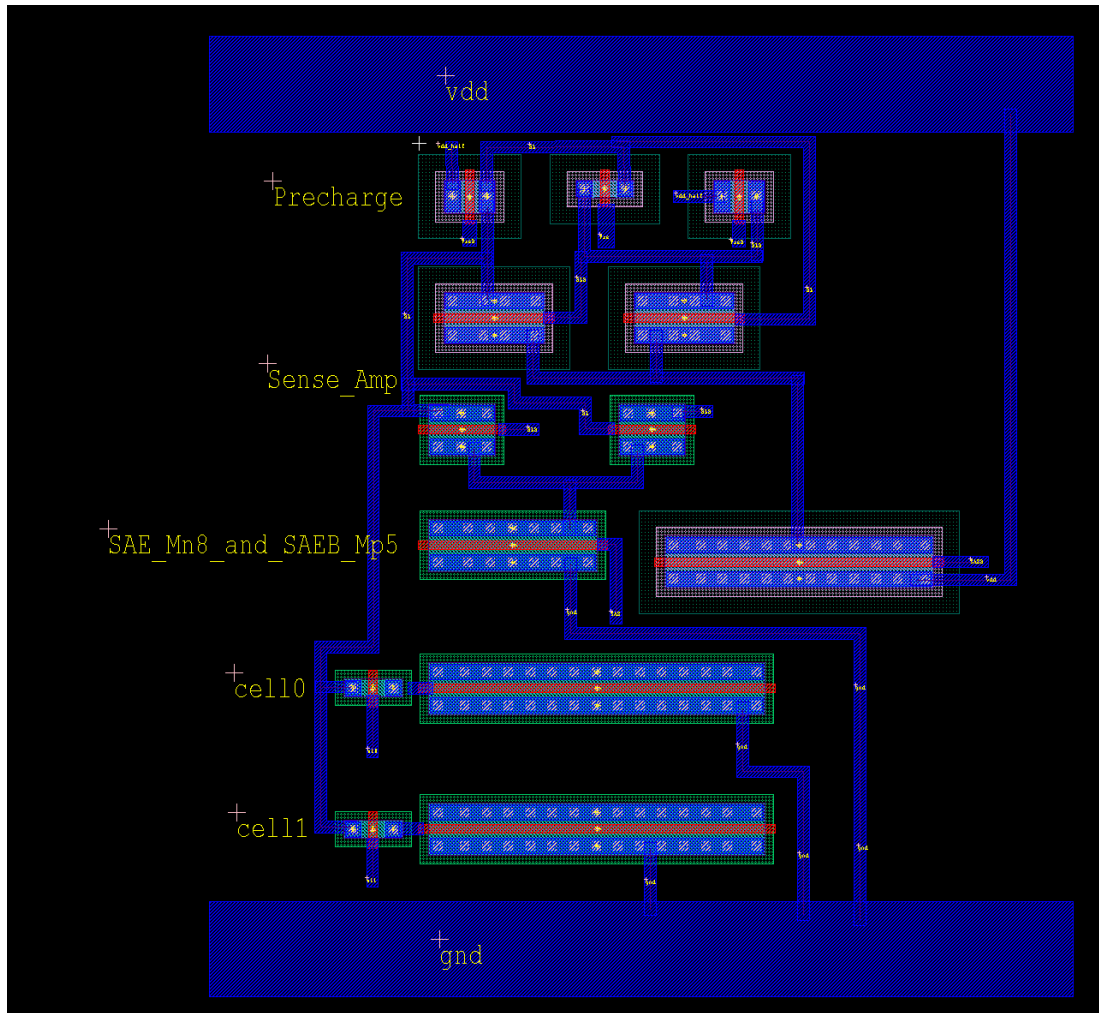


Figure 8: Physical layout in Siemens Tanner L-Edit. VDD rail at top, GND rail at bottom. Precharge, sense amplifier, SAE enable transistors, and two DRAM cells organized in functional rows. Metal1 routing and net labels visible throughout.

11. Design Decisions and Lessons Learned

The addition of SAE/SAEB required careful timing. The sense amplifier must not activate during precharge and can't activate before sufficient charge sharing differential has developed. For the WL0 cycle, a 2 ns delay between WL assertion and SAE activation (WL at 10 ns, SAE at 12.1 ns) allows the bitline differential to develop before the latch fires. For the WL1 cycle, the bitline differential begins recovering toward the precharge midpoint shortly after charge sharing peaks. SAE is fired at $t = 30.3$ ns, around 0.3 ns after WL1 assertion, to capture the differential at its maximum before recovery reduces it.

The precharge to $VDD/2$ is essential. With full VDD precharge, a stored logic 1 produces no differential because the storage node and bitline are at the same voltage. The $VDD/2$ supply connected to the precharge PMOS sources ensures both logic states create a detectable differential.

The VPWL source type was required for SAE timing because two pulse events per period (one per WL cycle) cannot be expressed as a single VPULSE. VPWL allows arbitrary waveforms and was used to define the SAE enable windows at 12-19 ns and 30.3-39 ns within each 40 ns period.

Separating PRE from PREB allowed independent control of the equalizer and the precharge transistors. This is the industry-standard approach and prevents the equalizer from fighting the precharge PMOS transistors during transitions.

Storage node voltage does not fully restore to VDD after a read. The sense amplifier restores charge through the access transistor, but the resistance of Mn4 limits the current and the 7 ns overlap window is insufficient for full restoration to VDD. In a real DRAM design, a dedicated write-back driver handles this but for simulation and time purposes I think this is acceptable.

Initial data values were set using SPICE initial conditions rather than a physical write operation. The .ic statements `.ic V(storage0)=2.5` and `.ic V(storage1)=0` instruct T-Spice to force the storage node voltages at the start of simulation, acting as a prior write without requiring a write driver. The 1T-1C cell topology supports write operations through the same access transistor used for reads in which driving BL to the desired voltage and asserting WL allows the storage capacitance/transistor to charge or discharge to match BL. However, implementing and verifying a write cycle wasn't included due to time limitations. The initial conditions approach still accurately represents the behavior of a cell that has already been written.

A dedicated $VDD/2$ DC source (1.25 V) was added to drive the sources of precharge transistors Mp1 and Mp2, rather than relying on the natural equalization behavior of the precharge circuit. In principle, with both PMOS sources tied to full VDD and Mn1 shorting BL to BLB, the two bitlines equalize to a midpoint voltage through charge sharing. In practice this midpoint depends on the matching of bitline parasitics and does not reliably land at exactly $VDD/2$. Using an explicit $VDD/2$ source guarantees a precise precharge level regardless of parasitic imbalance, and more clearly separates the precharge voltage from the supply rail in simulation. The explicit source also makes the design intent unambiguous when reading the schematic.

12. Conclusion

This project successfully designed, simulated, and laid out a two-cell 1T-1C DRAM array with a cross-coupled CMOS sense amplifier in Siemens Tanner S-Edit, T-Spice, and L-Edit using the Generic 250nm process at 2.5V supply. All primary objectives from the project contract were met.

The final circuit consists of 13 transistors implementing two complete DRAM storage cells, a cross-coupled latch sense amplifier with PMOS and NMOS halves, a sense amplifier enable circuit (SAE/SAEB), and a precharge block with independent Pre and PreB control. Both cells were verified in transient simulation storing opposite data values simultaneously, with Cell 0 initialized to logic 1 and Cell 1 initialized to logic 0. The sense amplifier correctly resolved BL and BLB to opposite rails for each cell during sequential read operations.

The measured sense margin of 57 mV at the moment of SAE activation exceeds the project target of 50 mV. Following sense amplifier activation, the bitlines resolved to a differential of approximately 1.5V, showing large differential amplification from an input signal under the threshold. Partial storage node refresh was observed and confirmed through simulation. Storage0 (logic 1) recovered to approximately 1.7 V after being depleted by charge sharing. This is consistent with the partial refresh limited by the NMOS threshold voltage drop across the access transistor. Storage1 (logic 0) was discharged to approximately 0.25 V following latch resolution, consistent with a correct logic 0 refresh where the latch drives BL low and transmits that state back through the access transistor. The incomplete restoration of storage0 is a known consequence of the NMOS access transistor limiting the maximum achievable storage voltage to approximately $V_{DD} - V_t$. This behavior is documented and consistent with published DRAM circuit theory.

The physical layout was completed in L-Edit using PDK-provided parameterized transistor cells with a top-to-bottom floorplan reflecting signal flow from VDD through the sense amplifier to the storage cells and GND. Metal1 routing and net labeling were applied throughout. DRC and LVS were not completed due to a Calibre license server configuration issue, which is noted as a limitation of the current implementation environment rather than a design deficiency.

Overall, the project provided direct experience with the full custom IC design flow from circuit specification through schematic entry, transient simulation, timing analysis, and physical layout, using the same commercial toolchain used in industry VLSI development.